



BITS Pilani

Essential Mathematics for Datascience

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DSEZC411 Essential Mathematics for Data Science

Lecture 1

What is linear algebra?



- ▶ Linear algebra is the study of vectors and rules to manipulate vectors.
- ▶ Vectors are not only the familiar geometric vectors from high school (points in 2D/3D space)
- ▶ Vectors can be any special objects which can be added together and multiplied by scalar values to produce another object of the same kind.
- ▶ For example, polynomials can also be treated as vectors.
- ▶ We shall deal with vectors in the space $\mathbb{R}^n$



A vector $\mathbf{x} \in \mathbb{R}^n$ is a tuple of n real elements. The i^{th} entry of $\mathbf{x}$ is referred to as $\mathbf{x}_i$.

Example: $\mathbf{x} = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 40 \\ 50 \end{bmatrix}$

- ▶ Two common operations on vectors are vector addition and scalar multiplication.

- ▶ $\mathbf{x}_1 + \mathbf{x}_2 = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix} + \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$

- ▶ $3.\mathbf{x}_1 = 3. \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}$

Dot Product of vectors in $\mathbb{R}^n$



A dot product between vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ is defined as follows:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n \mathbf{x}_i \mathbf{y}_i$$

Example: Let $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, then $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n \mathbf{x}_i \mathbf{y}_i = 0$

An (m, n) matrix A is a mn tuple of elements a_{ij} arranged in m rows and n columns as below:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- Note that row and column vectors are also matrices - a row vector is a $(1, n)$ matrix and a column vector is a $(m, 1)$ matrix.
- $\mathbb{R}^{m \times n}$ is the set of all (m, n) matrices consisting of real numbers as elements



The sum of two matrices $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times n}$ is defined as an element-wise sum of the elements of the two matrices:

$$A = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & & & \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

Elementary Row Operations on a matrix



There are three fundamental operations on a matrix. They are called elementary row operations. There are as follows:

1. Exchange of two rows
2. Multiplying a row by a nonzero constant number
3. Adding a multiple of one row to the current row

Elementary Row Operation

Exchange of two rows



Example: $\mathbf{R}_1 \leftarrow \mathbf{R}_3$: Swap rows 1 and 3 in the matrix to get

$$\begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ 1 & -2 & 1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -3 & 4 & -1 \end{bmatrix}$$

Elementary Row Operation

Scaling of a row



Example: $\mathbf{R}_4 \leftarrow 4\mathbf{R}_4$:

$$\begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 0 & -12 & 16 & -4 \end{bmatrix}$$

Elementary Row Operation

Adding a multiple of one row to the current row



Example: $\mathbf{R}_2 \leftarrow \mathbf{R}_2 - 4\mathbf{R}_1$,

$$\begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 0 & -12 & 4 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 0 & -12 & 4 & -4 \end{bmatrix}$$

Definition: Row Echelon form



Pivot entry in a row: the first non-zero element in a given row is called the pivot.

The row echelon form property are:

- ▶ All rows that contain at least one nonzero element are on top of rows that contain only zeros. All rows that contain only zeros are at the bottom of the matrix.
- ▶ The pivot in a row is always to the right of the pivot in the row above it. The positions of the pivots in the non-zero rows give rise to a staircase pattern.

Example: Row Echelon Form



$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

Example: Row Echelon Form



Consider a matrix, $M = \begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ -2 & 4 & -2 & -1 & 4 & -3 \\ 1 & -2 & 0 & -3 & 4 & -1 \end{bmatrix}$

$$\rightarrow \begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & -1 & -2 & 3 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 & -3 & 2 \\ 0 & 0 & 0 & -3 & 6 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



We can convert the row-echelon form into a simpler form called the reduced row-echelon form.

- ▶ In reduced row-echelon form, every pivot is equal to 1.
- ▶ The pivot is the only non-zero entry in its column.

Example

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & 2 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

A system of linear equations



Systems of linear equations form a central part of linear algebra. For example, consider a system of linear equations:

$$\begin{aligned}x_1 + x_2 + x_3 &= 2 \\ 2x_1 + x_2 + x_3 &= 3\end{aligned}$$

It can be restated as:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Tools of linear algebra can be used to solve such problems.

Homogeneous linear system



1. If b is zero vector, then $Ax = b$ is called a homogeneous linear system

$$\begin{bmatrix} 2 & 3 & 5 \\ 4 & -2 & -7 \\ 9 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

2. If b is not a zero vector, then $Ax = b$ is called a non homogeneous linear system

Consider the following system of equations:

$$2x_1 + 3x_2 + 5x_3 = 1$$

$$4x_1 - 2x_2 - 7x_3 = 8$$

$$9x_1 + 5x_2 - 3x_3 = 2$$

In matrix terms we can write this set of equations as

$$\begin{bmatrix} 2 & 3 & 5 \\ 4 & -2 & -7 \\ 9 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ 2 \end{bmatrix}$$

- We see that a system of equations can be expressed as $Ax = b$ where $A \in \mathbb{R}^{m \times n}$, x is a $n \times 1$ matrix and b is a $m \times 1$ matrix.



Consider the linear system

$$-2x_1 + 4x_2 - 2x_3 - x_4 + 4x_5 = -3$$

$$4x_1 - 8x_2 + 3x_3 - 3x_4 + x_5 = 2$$

$$x_1 - 2x_2 + x_3 - x_4 + x_5 = 0$$

$$x_1 - 2x_2 - 3x_4 + 4x_5 = -1$$

$$\begin{bmatrix} -2 & 4 & -2 & -1 & 4 & -3 \\ 4 & -8 & 3 & -3 & 1 & 2 \\ 1 & -2 & 1 & -1 & 1 & 0 \\ 1 & -2 & 0 & -3 & 4 & -1 \end{bmatrix}$$

This matrix is called the augmented matrix.

Row Echelon Form of Augmented Matrix



$$\begin{bmatrix} 1 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 3 & -2 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

We can revert to the set of equations represented by the augmented matrix as follows. These equations are equivalent to the original set of equations:

$$\begin{aligned} x_1 - 2x_2 + x_3 - x_4 + x_5 &= 0 \\ x_3 - x_4 + 3x_5 &= -2 \\ x_4 - 2x_5 &= 1 \\ 0 &= 0 \end{aligned}$$

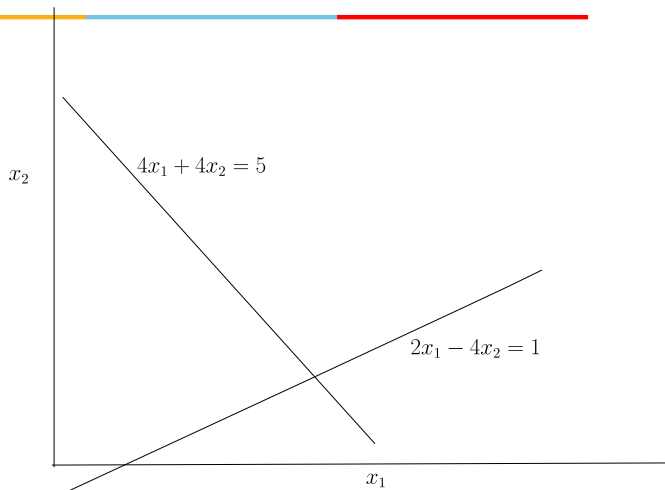
Does a linear system always have solutions?



A linear system has zero, one or infinitely many solutions

1. The solution to linear equations can be given a geometrical interpretation.
2. When equations are given in terms of two variables, the solution to two equations in two variables could be a point (unique solution), a line (infinite solutions) or no solution (parallel lines).

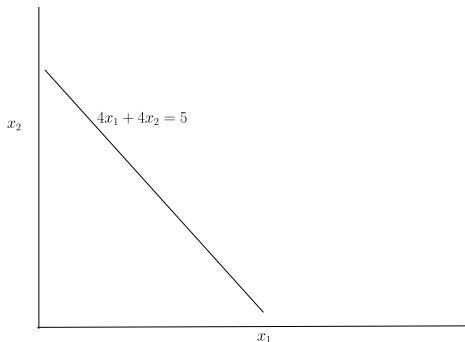
One Solution



Infinite Solutions



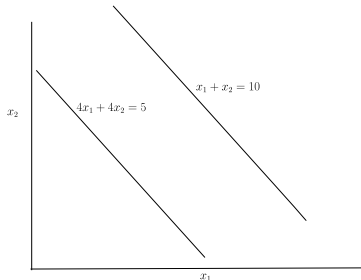
In the second case both constraints are the same, so there are an infinite number of solutions:



No Solution



In the third case the constraints are mutually incompatible, so there is no assignment to x_1, x_2 which satisfies both constraints. The graph of both constraints shows a pair of parallel lines:



Equations in higher dimensions



- ▶ In 3D each constraint is a plane and the intersection of two planes is a line.
- ▶ The intersection of the third plane with the first two planes will be a point on the line in case of a unique solution,
- ▶ or it may lead to pairs of parallel lines (constraint 1 intersection constraint 2 gives one line, constraint 1 intersection constraint 3 gives a parallel line, which means there is no solution.
- ▶ or all three planes may intersect in the same line which means infinite solutions.

Gaussian Elimination

Algorithm to solve linear equations



- ▶ The key idea is to take a complex looking matrix and transform it using elementary row operations to a simple looking matrix like the one we just handled, for which solutions could be obtained essentially by inspection.
- ▶ To make this work we need to preserve solutions of the original system of equations, i.e ensure that elementary transformations of the original matrix do not change its solutions.

Example $Ax = b$



$$\begin{bmatrix} 1 & 3 & 2 & 0 & 21 \\ 0 & 0 & 2 & 0 & 18 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix}$$

The Augmented matrix is

$$\left[\begin{array}{ccccc|c} 1 & 3 & 2 & 0 & 21 & 5 \\ 0 & 0 & 2 & 0 & 18 & 4 \\ 0 & 0 & 0 & 1 & -4 & 0 \end{array} \right]$$

Particular Solution of $Ax=b$



$$\begin{bmatrix} 1 & 3 & 2 & 0 & 21 & | & 5 \\ 0 & 0 & 2 & 0 & 18 & | & 4 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} 1 & 3 & 2 & 0 & 21 & | & 5 \\ 0 & 0 & 1 & 0 & 9 & | & 2 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 3 & | & 1 \\ 0 & 0 & 1 & 0 & 9 & | & 2 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (3)$$

The particular solution is $x_p = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}$

Solution of $Ax=0$



$$\begin{bmatrix} 1 & 3 & 2 & 0 & 21 & | & 0 \\ 0 & 0 & 2 & 0 & 18 & | & 0 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} 1 & 3 & 2 & 0 & 21 & | & 0 \\ 0 & 0 & 1 & 0 & 9 & | & 0 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (5)$$

$$\begin{bmatrix} 1 & 3 & 0 & 0 & 3 & | & 0 \\ 0 & 0 & 1 & 0 & 9 & | & 0 \\ 0 & 0 & 0 & 1 & -4 & | & 0 \end{bmatrix} \quad (6)$$

The solution of homogeneous system is $x_h = \lambda_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -3 \\ 0 \\ -9 \\ 4 \\ 1 \end{bmatrix}$

General Solution of $\mathbf{Ax} = \mathbf{b}$



The general solution of $Ax = b$ is $x_g = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} -3 \\ 0 \\ -9 \\ 4 \\ 1 \end{bmatrix}$

Calculating matrix inverse



- ▶ If $\mathbf{A}$ is invertible, it means that $\mathbf{A}^{-1}$ exists such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$
- ▶ In such a case the row-reduced echelon form of $\mathbf{A}$ is $\mathbf{I}_n$, i.e every column is a pivot column where the pivot is 1.

Gauss Jordan Algorithm



Find the inverse of $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$